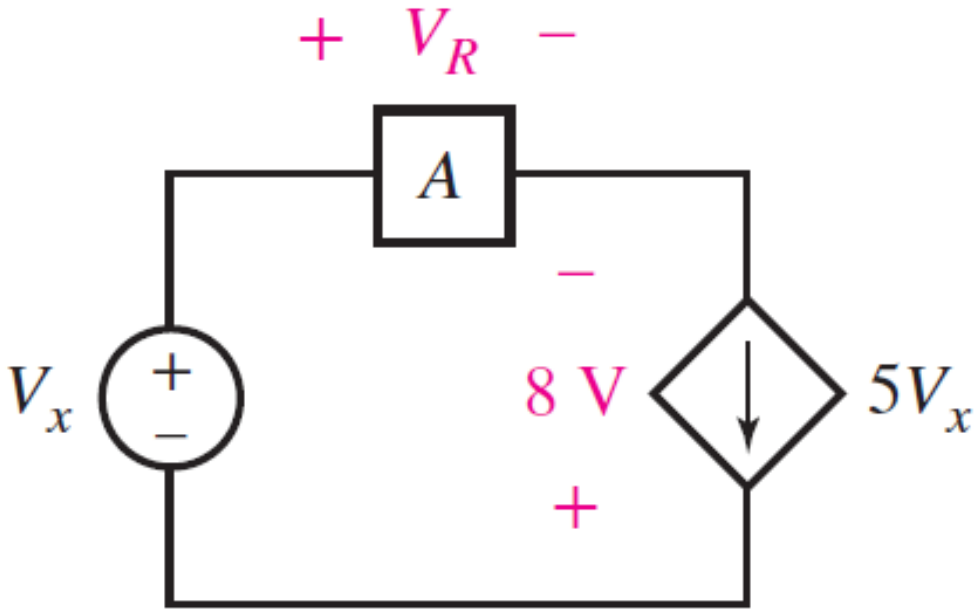
Q1: Refer to the circuit below, while noting that the same current flows through each element. The voltage-controlled dependent source provides a current which is 5 times as large as the voltage Vx.

(a) For $V_R = 100$ V and $V_X = 92$ V, determine the power absorbed by each element.
(b) Verify that the algebraic sum of the absorbed powers is equal to zero.



Q2: Sketch the power absorbed by a 100 Ω resistor as a function of voltage over
the range $-2\ \text{V} \leq V_{resistor} \leq +2\ \text{V}$.